# JAMAL MOHAMEDAIN

Mechanical & Mechatronics Engineer | EIT Certified

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LinkedIn Profile | Portfolio Website
Nationalities: French | Egyptian | Sudanese
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Mechanical and mechatronics engineer (B.Eng., EIT) who designs, builds and commissions: CATIA V5 and SolidWorks design with Ansys FEA/CFD, robotic and automated systems (Siemens PLC, ESP32), production-line commissioning, and a preventive maintenance programme that cut unplanned downtime by 48% across seven lines.

## EXPERIENCE

### Operations Lead
Feb 2026 – May 2026

**KBM Technologies** | Riyadh, Saudi Arabia

- Managed development of the Caters app from planning to delivery, tracking progress against deadlines and re-planning when priorities or resources changed.
- Coordinated a 5-person team of developers and stakeholders: ran progress follow-ups, raised schedule risks early, and aligned decisions between technical and business sides.
- Set up the project's technical infrastructure: GitHub organisation and access structure, Google Workspace domain configuration, and Google Cloud IAM permissions.

### Production & Maintenance Intern
Jul 2024 – Oct 2024

**L'Oréal Cairo Plant** | Cairo, Egypt

- Developed an Autonomous Preventive Maintenance (APM) program across seven production lines through detailed mechanical inspection of filler machines (camshaft assemblies, chain and belt drives, moving parts).
- Documented findings as structured inspection and cleaning procedures issued automatically via MaintainX CMMS, and tracked results, reducing unplanned downtime by 48%.
- Participated in the commissioning of a new production line, including LOTO (Lockout/Tagout) procedures and safety protocols.
- Led the renovation of the weighing room: specified a shaking table and carts that removed heavy manual labour from batch preparation and designed a piping system for the room (quick-connect fittings; not implemented, as installation exceeded the internship timeline).

### Production & Maintenance Intern
Aug 2020 – Sep 2020

**MEGA Food Industries | Sudanese PEPSI Bottler** | Khartoum, Sudan

- Gained foundational industrial experience in safety, hygiene standards, machine operation, and maintenance procedures in a high-volume FMCG manufacturing environment.
- Supported production planning and machine maintenance tasks across bottling and packaging lines.

## EDUCATION

### B.Eng. in Mechanical and Mechatronics Engineering
Sep 2022 – Jul 2025

**University of Hertfordshire (UH) – Hosted by GAF** | Cairo, Egypt

- Graduation project: Wall-Climbing Robot for silo inspection and maintenance.

### Mechanical Engineering (First Year Completed)
Sep 2019 – Sep 2022

**University of Khartoum** | Khartoum, Sudan

- Completed first year and was promoted to second year. Studies were repeatedly interrupted by university closures (COVID-19 and the October 2021 coup) before transferring to UH.

### IGCSE & AS-Level
Sep 2016 – Jun 2019

**Manarat International School** | Jeddah, KSA

- Completed IGCSE and AS-Level certifications across a 3-year period.

## NOTABLE PROJECTS

### Wall-Climbing Robot [Graduation Project]

Individual, self-proposed graduation project: sole designer of a wall-climbing robot for silo inspection, delivered within the university's £75 budget cap. Modelled all parts and the full assembly in CATIA V5 (Part Design, Assembly Design, Generative Shape Design, Drafting), including a propeller built from a NACA 0015 profile. Simulated its airflow in CFD using Ansys Workbench (Fluent CFD), predicting about 5 N of thrust for a 500 g robot. Produced 2D drawings for laser-cut parts and 3D printed complex-shaped ones. Traced an adhesion failure to air leaking through the ground clearance and fixed it with a compliant foam seal. When the printed propellers

failed under load, simulated a commercial fan design in CFD and replaced them with a commercial fan blade, adapting the housing to fit. Also handled electronics and control.

**Soft Robotic Gripper for DOBOT** [Control Systems]

Designed the gripper from scratch in CATIA V5 instead of using the provided design, including its mould. Cast the gripper in silicone using a 3D-printed PLA mould. Controlled it with PID and Adaptive PID algorithms for precision manipulation.

**Seaport Crane Gantry System** [Control Systems]

Designed in CATIA V5 (Part Design, Assembly Design, Drafting) and built a 3-axis Cartesian gantry rig modelling seaport crane kinematics using aluminum extrusion, encoded DC motors, and an ESP32 microcontroller. Implemented and benchmarked five control algorithms (PID, Adaptive PID, Fuzzy-Like PID, Self-Tuning Fuzzy, MFAC) via MATLAB over Wi-Fi TCP, with Adaptive PID achieving best tracking performance through real-time gain scheduling.

**PLC Industrial Palletizing Simulation** [Siemens TIA Portal]

Designed an automated industrial palletizing line using a Siemens PLC, programmed with Ladder Diagrams in Siemens TIA Portal and simulated in Factory IO.

**HUSKY UGV Mobile Robot** [Autonomous Systems]

Team lead. Designed the chassis in CATIA V5 to match the original dimensions, with motor mounts and access to the electronics. Tested forward and inverse kinematics and PID speed control in Gazebo before assembly.

## CERTIFICATIONS

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| **Engineer in Training (EIT) - NCEES FE Mechanical** | April 2026 |
| Google — Professional Certificate in Business Intelligence | June 2024 |
| Google — Professional Certificate in Data Analytics | May 2024 |
| UH — Finite Element Analysis Workshop | August 2024 |
| UH — Embedded Systems Design Workshop | July 2023 |
| Jet Solar — Manufacturing & Commissioning of Solar Panels | August 2023 |
| Arab American Vehicle Co. — Car Assembly, QC & Maintenance | June 2023 |
| AOI — CNC Operation | March 2023 |

## SKILLS

| Technical | Tools & Software | Languages |
|---|---|---|
| Mechanical Design (CAD) | SolidWorks, CATIA V5 (Part Design, Assembly Design, GSD, Drafting, Basic FEA), SketchUp | Arabic — Native |
| PLC Programming | Siemens TIA Portal, Factory IO | English — Fluent |
| Embedded Systems | Arduino, ESP32, C++, Python | French — Beginner |
| FEA & CFD | Ansys Workbench (Mechanical, Fluent CFD) | |
| 3D Printing & Prototyping | UltiMaker Cura, laser cutting, silicone moulding | |
| Circuits, Control & Simulation | Proteus, MATLAB, Simulink | |
| Commissioning & LOTO | LOTO procedures, MaintainX CMMS | |
| Analysis & Data | Python, MATLAB, MS Excel, SQL, Google BI | |